

Question

Electrolysis of an aqueous solution with a certain concentration of HF and AgHF_2 yields a crystalline substance **X** when the current density is high. **X** exhibits paramagnetism and good conductivity. Experimental results confirm that **X** has a face-centered cubic lattice structure. The structural description of **X** is as follows:

The O_8 small cubes composed of oxygen occupy all the vertices and face centers of the unit cell, while randomly oriented HF_2^- occupy all the edge centers and body centers. Each small cube has a Ag(I) at its center, and the remaining silver atoms are distributed between the cubes, with one silver atom located at the midpoint of the line connecting the centers of any two nearest O_8 small cubes.

Which of the following statements is correct:

- A.** The simplest chemical formula of **X** contains 8 metal atoms
- B.** The structural unit of **X** contains 36 atoms
- C.** The Ag between the cubes is sp^3 hybridized.
- D.** In a conventional unit cell, there are 4 tetrahedral voids composed of O_8 .
- E.** Crystal **X** is deposited at the cathode
- F.** Given that $a = 982 \text{ pm}$, the density of this crystal is greater than 8 g/cm^3 .
- G.** None of the above options is correct.

Answer

Correct Answer: **G**

Detailed Explanation

The O_8 small cubes occupy all the vertices and face centers of the unit cell, so there are 4 O_8 small cubes in a unit cell.

HF_2^- is randomly oriented, occupying all the edge centers and body center of the unit cell, and similarly, there are 4 HF_2^- in a unit cell.

CHECKPOINT

1 PTS

A unit cell contains 4 O_8 small cubes and 4 HF_2^-

The midpoint of the line connecting the centers of any two nearest O_8 small cubes should coincide with the midpoints of the lines connecting all vertices and face centers, as well as the midpoints of the lines connecting two face centers. Thus, there are four Ag on each face and 12 Ag inside the crystal, totaling $4 \times 6/2 + 4 + 12 = 28$ Ag in the crystal.

CHECKPOINT

1 PTS

The midpoint of the line connecting the centers of the nearest O_8 small cubes coincides with the midpoints of the lines connecting all vertices and face centers, as well as the midpoints of the lines connecting two face centers

CHECKPOINT

1 PTS

The crystal contains a total of $4 \times 6/2 + 4 + 12 = 28$ Ag

Since this crystal has a face-centered cubic lattice, a primitive unit cell contains 4 structural motifs. The chemical formula in one unit cell is $(\text{O}_8)_4(\text{HF}_2^-)_4\text{Ag}_{28}$, so the structural motif is $(\text{O}_8)(\text{HF}_2^-)\text{Ag}_7$, which is also the simplest chemical formula. Options B and A are both incorrect.

CHECKPOINT

1 PTS

The structural motif is $(\text{O}_8)(\text{HF}_2^-)\text{Ag}_7$

Ag(I) is located between the O_8 cubes. In the simplest chemical formula, only one exists. Since the crystal is paramagnetic and Ag(I) has no unpaired electrons, the Ag between the cubes has unpaired electrons and a valence of +2/+3, resulting in dsp^2 hybridization. Option C is incorrect.

CHECKPOINT

1 PTS

The Ag between the cubes has unpaired electrons and exhibits dsp^2 hybridization

In a primitive unit cell, O_8 occupies the vertices and face centers, which are actually the positions of cubic close packing. Therefore, O_8 forms 8 regular tetrahedral voids. Option D is incorrect.

CHECKPOINT

1 PTS

The oxygen atoms are actually in cubic close packing positions, forming 8 regular tetrahedral voids

Considering the simplest chemical formula, the apparent oxidation state of silver is +17/7, indicating an oxidation process. Thus, the crystal should precipitate at the anode. Option E is incorrect.

CHECKPOINT

1 PTS

The formation of the crystal is an oxidation process, and the crystal should precipitate at the anode

The density calculated based on the unit cell parameters and the simplest chemical formula is 6.47 g/cm^3 . Option F is incorrect.

CHECKPOINT

1 PTS

The density is 6.47 g/cm^3

In conclusion, options A-F are all incorrect, and option G is correct.